MAX PLANCK INSTITUTE
OF ANIMAL BEHAVIOR

ECOLOGY of
ANIMAL
SOCIETIES

Communication
& Coordination
Across Scales

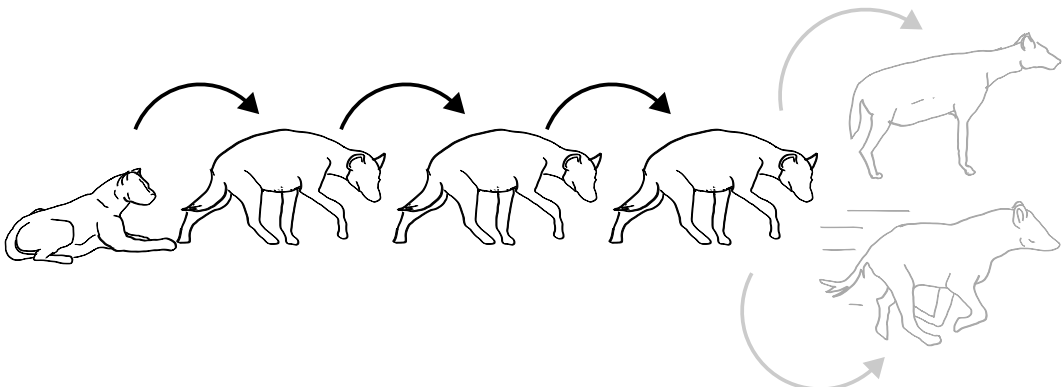


The Behavioural Observatory

Cross-species comparison of behavioural sequences in the wild

Pranav Minasandra

When to change your behavioural state?



When to change your behavioural state?

- Animal behaviour unfolds as a sequence of behavioural decisions.

When to change your behavioural state?

- Animal behaviour unfolds as a sequence of behavioural decisions.
- What does the algorithm that generates this set of behaviours look like?

When to change your behavioural state?

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- What does the **algorithm** that generates this set of behaviours look like?

What is a behavioural algorithm?

What is a behavioural algorithm?

How an animal decides behaviour in the next instant.

What is a behavioural algorithm?

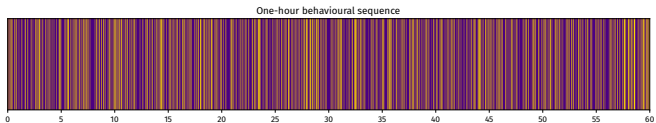
How an animal decides behaviour in the next instant.

- E.g., actions are chosen randomly each instant.

What is a behavioural algorithm?

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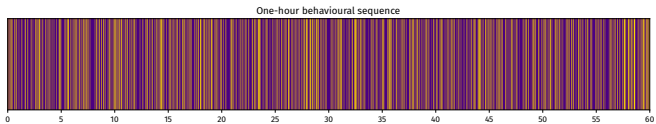
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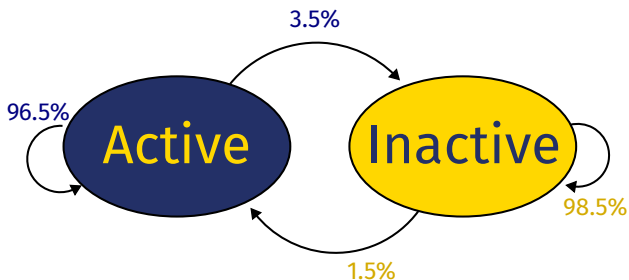
- E.g., actions are chosen randomly each instant.
- Current-state-dependent transition probabilities



What is a behavioural algorithm?

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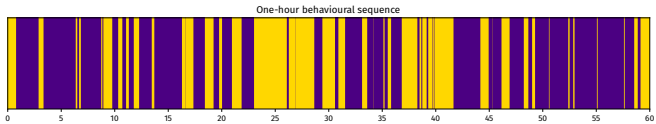
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What is a behavioural algorithm?

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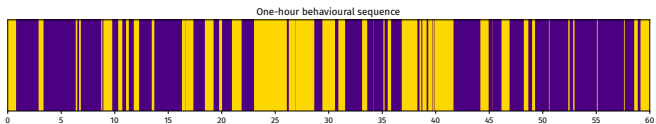
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What is a behavioural algorithm?

How an animal decides behaviour in the next instant.

- E.g., actions are chosen randomly each instant.
- Current-state-dependent transition probabilities, i.e., a Markov model.
- Or something more complex.



Observing natural behaviour with accelerometers

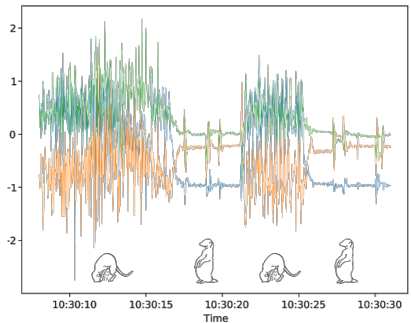
Observing natural behaviour with accelerometers

- Behavioural data from three species.



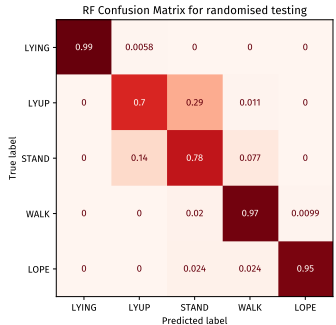
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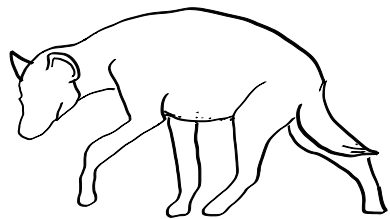
- Behavioural data from three species.
- Recorded using accelerometer collars.
- Behavioural sequences inferred using machine learning methods.



Exploring behavioural sequences

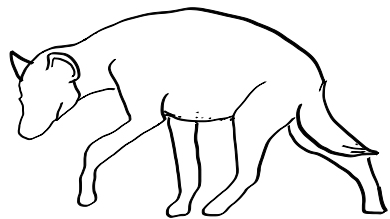
Exploring behavioural sequences

- A hyena has just started to walk.



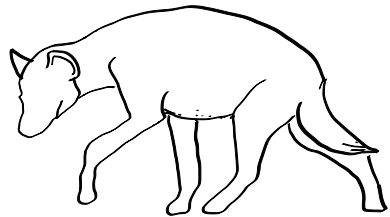
Exploring behavioural sequences

- A hyena has just started to walk.
- Consider the probability that it stops and does something else in the next instant.



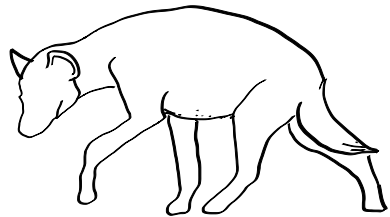
Exploring behavioural sequences

- A hyena has just started to walk.
- Consider the probability that it stops and does something else in the next instant.
- How does this probability change the longer the hyena walks?



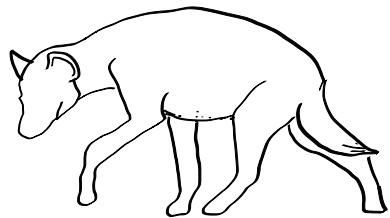
Exploring behavioural sequences

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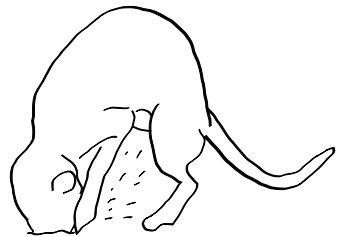
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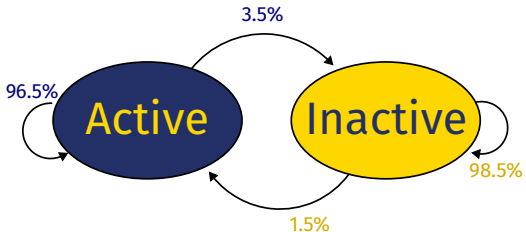


Hazard functions

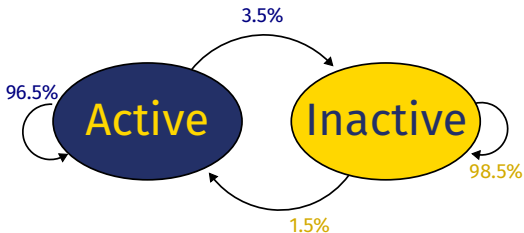
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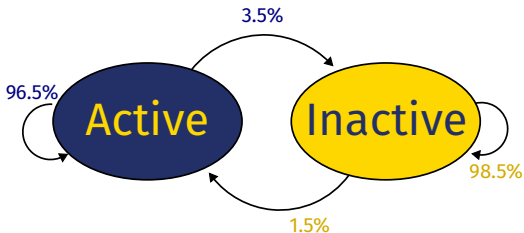


Hazard functions



As these transition probabilities are fixed, the hazard function remains flat.

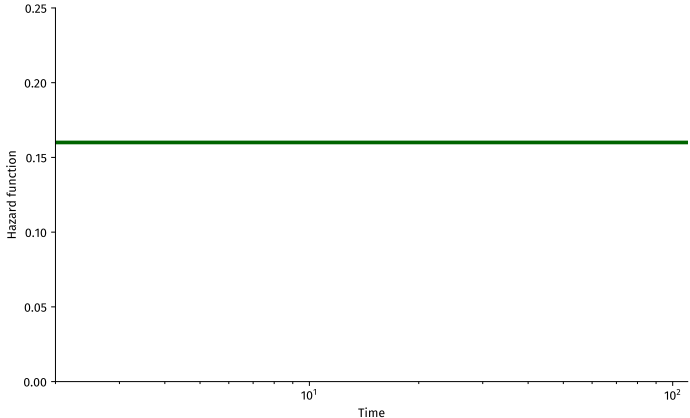
Hazard functions



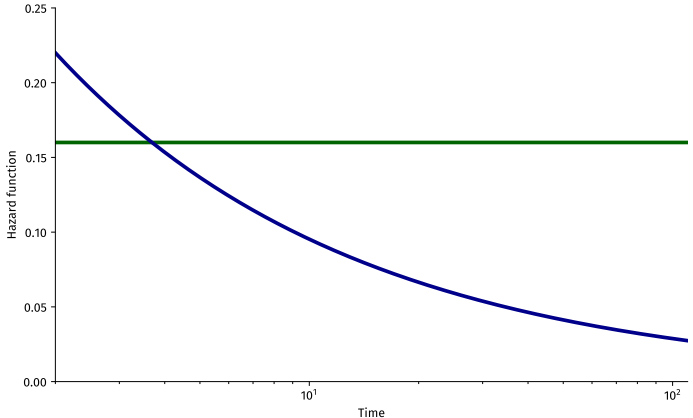
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Hazard functions—hypotheses

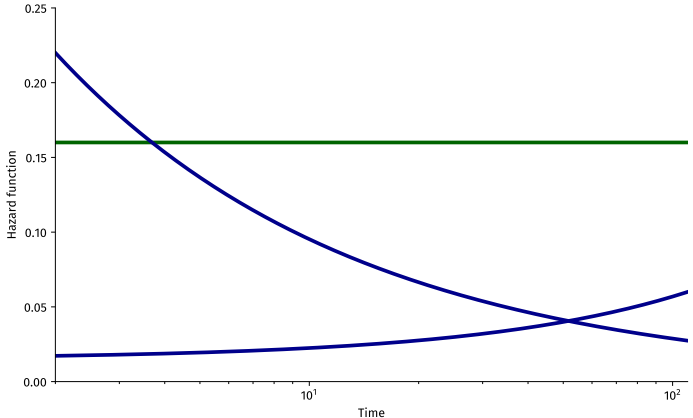
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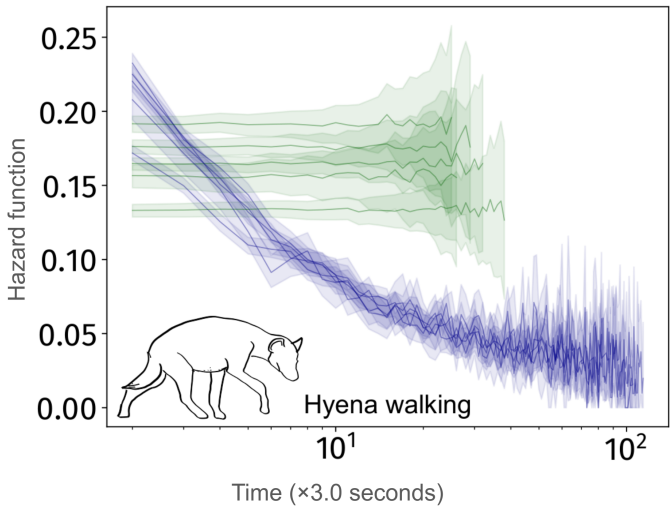
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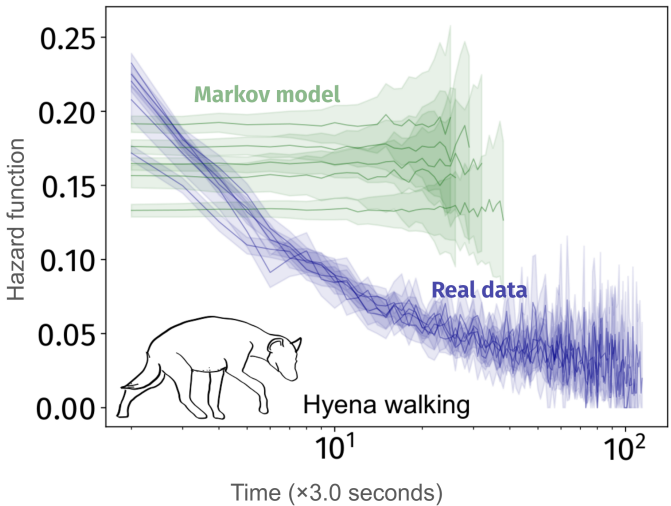
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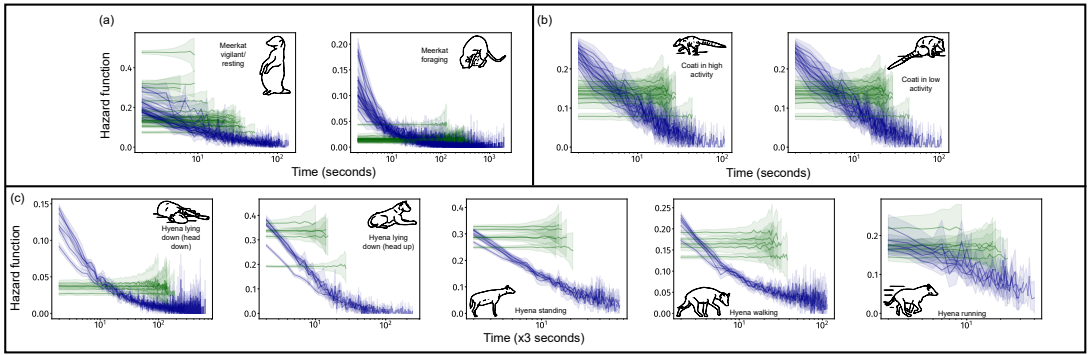
Hazard Functions—In reality



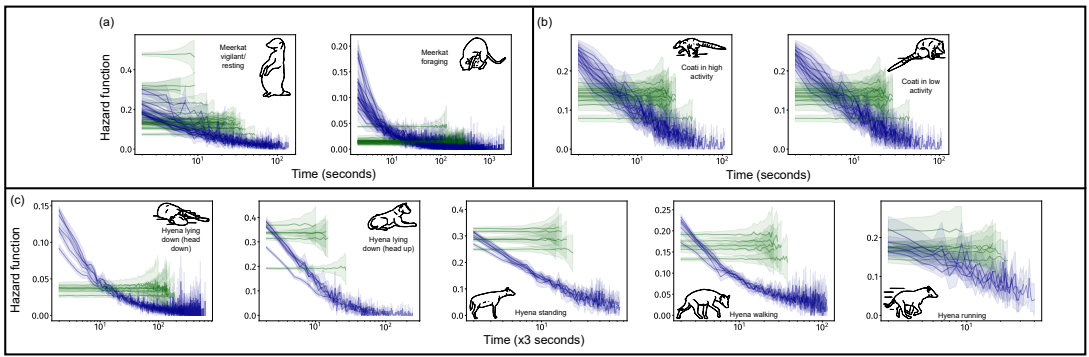
Hazard Functions—In reality



Hazard Functions—In reality



Hazard Functions—In reality



The longer an animal does something, the less likely it is to switch to do something else.

PNAS RESEARCH ARTICLE | ECOLOGY OPEN ACCESS

Behavioral sequences across multiple animal species in the wild share common structural features

Pranav Minasandra [a,b,c,d,1], Emily M. Grout [a,b,c,d,e], Katrina Brock [a], Margaret C. Crofoot [a,b,d,e], Vlad Demartsev [a,b,d,f], Andrew S. Gersick [g], Ben T. Hirsch [e,h], Kay E. Holekamp [i,j], Lily Johnson-Ulrich [f,k], Amlan Nayak [a,b,l], Josué Ortega [a,e], Marie A. Roch [m], Eli D. Strauss [a,b,d,i,j], and Ariana Strandburg-Peshkin [a,b,d,f]

Edited by Gene Robinson, University of Illinois at Urbana-Champaign Institute for Genomic Biology, Urbana, IL; received February 21, 2025; accepted April 18, 2025

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One of three shared results across species.

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- Are these universal phenomena?

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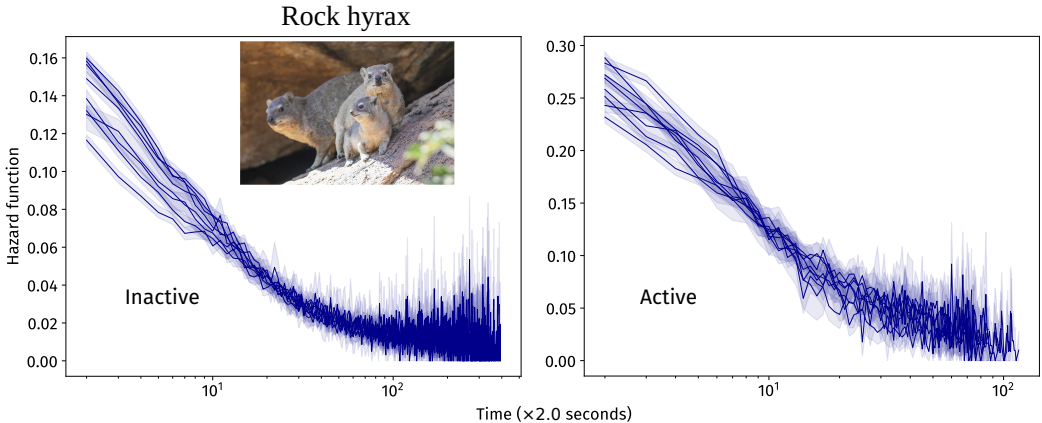
- Are these universal phenomena?
- Why do they happen?

Possibly quite widespread

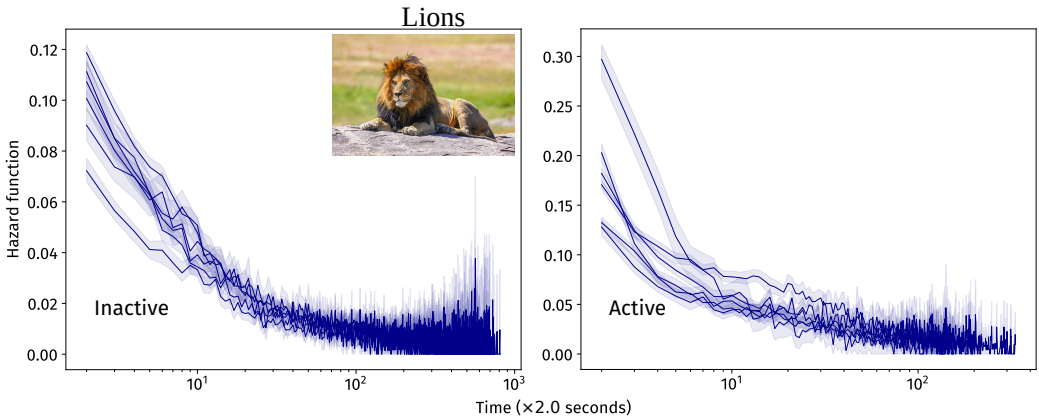
Possibly quite widespread

Cross-species behaviour delineation system

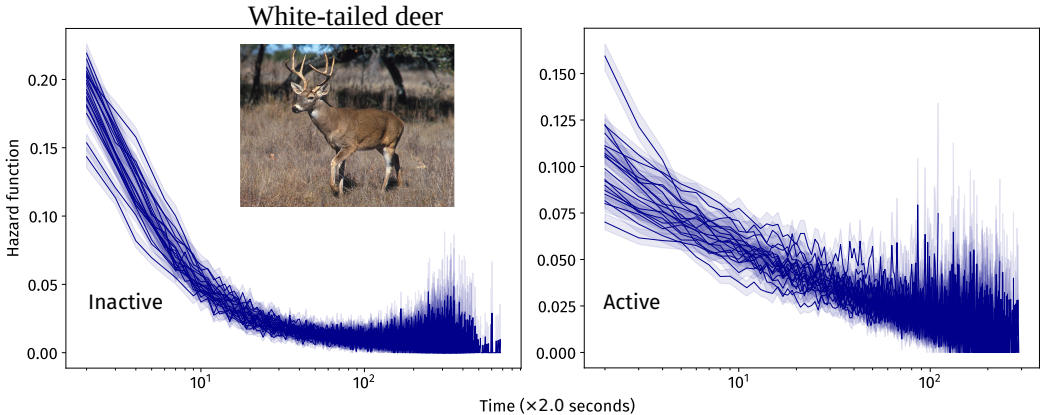
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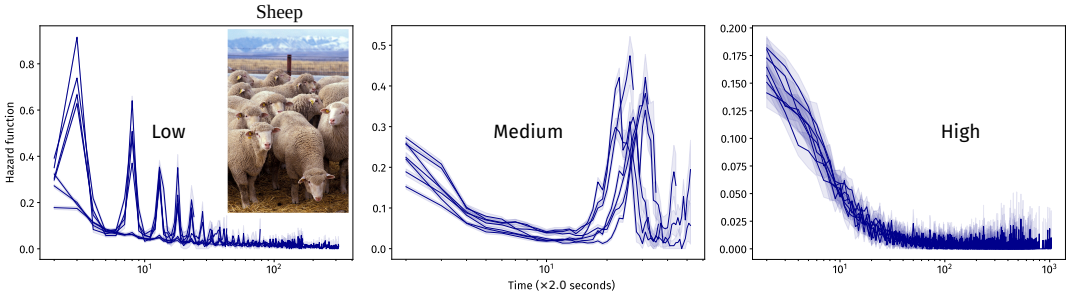
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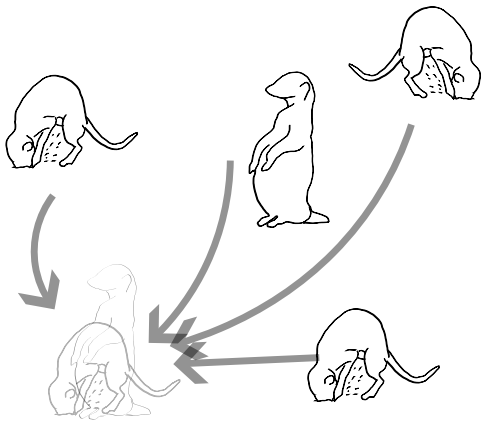


But what if all this is an artefact of the methods used?

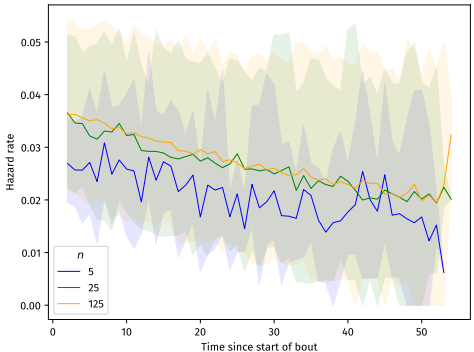

Sheep
Low
Medium
High
Hazard function
Time (×2.0 seconds)

Why does this happen?—Simulations

Why does this happen?—Simulations

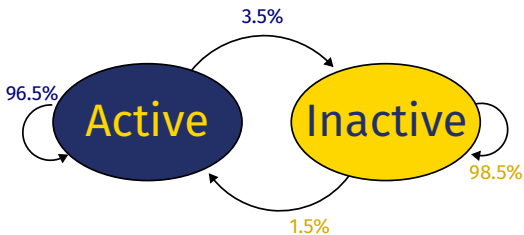


Why does this happen?—Simulations



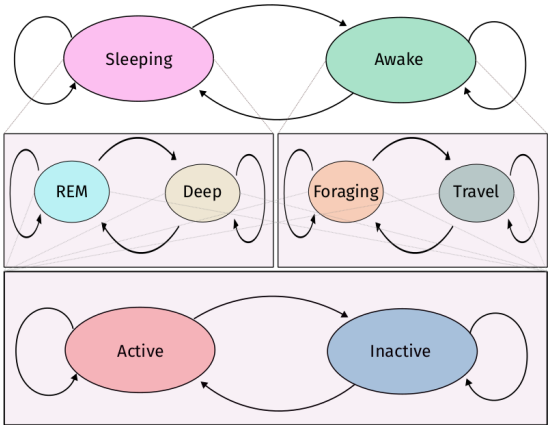
- Interacting animals tend to produce the same result.

Why does this happen?—Simulations



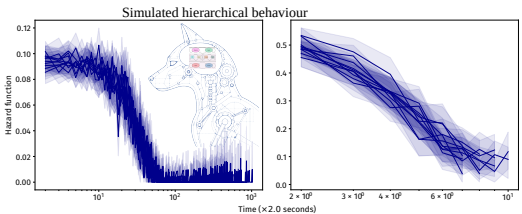
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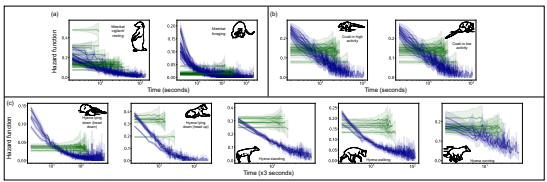
Why does this happen?—Simulations



- Interacting animals tend to produce the same result.
- Hierarchical behaviour tends to produce the same result.

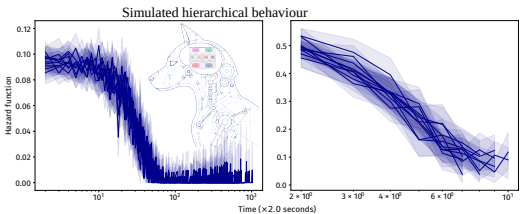
Take home messages

Take home messages



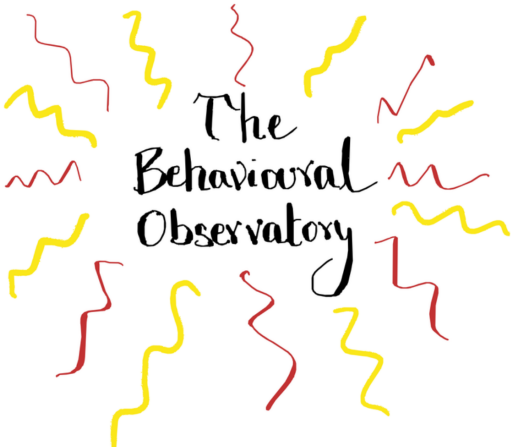
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Take home messages



- A quite widespread behavioural algorithm.
- Simulations allow us to test candidate behavioural algorithms.

Take home messages



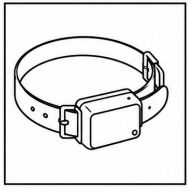
- A quite widespread behavioural algorithm.
- Simulations allow us to test candidate behavioural algorithms.
- We welcome new collaborations.

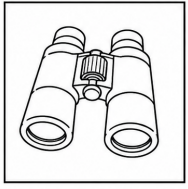
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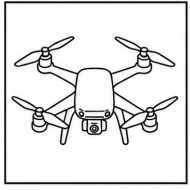
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[This list is incomplete. You can help by expanding it.]

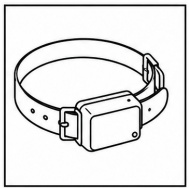
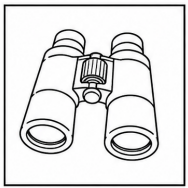
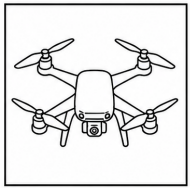
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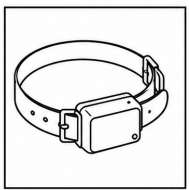
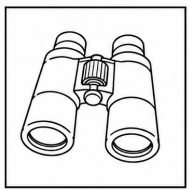
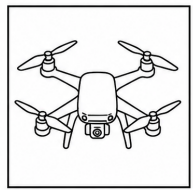


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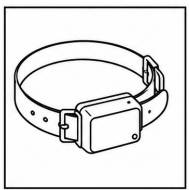
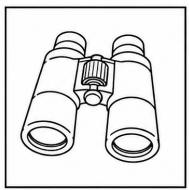
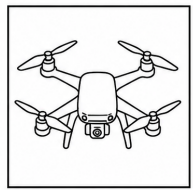
- Behavioural recordings that are continuous ,

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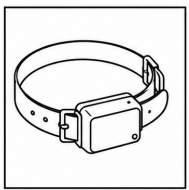
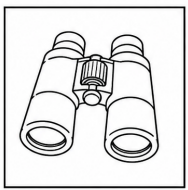
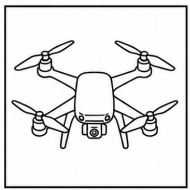
- Behavioural recordings that are continuous, multi-day,

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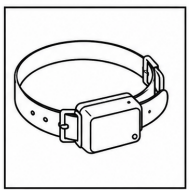
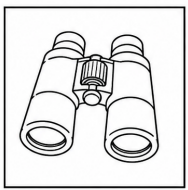
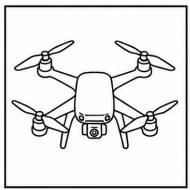
- Behavioural recordings that are continuous, multi-day, preserve individual identity.

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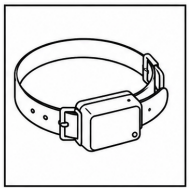
- Behavioural recordings that are continuous, multi-day, preserve individual identity.
- $\approx$ 10 or more individuals.

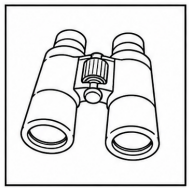
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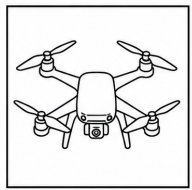




- Behavioural recordings that are continuous, multi-day, preserve individual identity.
- $\approx$ 10 or more individuals.
- $\approx$ 10 or more days of data.

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- Behavioural recordings that are continuous, multi-day, preserve individual identity.
- ≈ 10 or more individuals.
- ≈ 10 or more days of data.

pminasandra@ab.mpg.de